SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

---

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.
(b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.
(b) Given $s = clog(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4

Intensity	Frequency
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

$$[10\ 20\ 30\ 20\ 30\ 40\ 30\ 40\ 50\ ]$$

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$$[[0\ \ -1\quad 0\}, \{-1\quad 4\ \ -1],\ \ \{\ 0\ \ -1\quad 0\ ]]$$

to the matrix:

$$[10\ \ 10\ \ 10,\ 10\ \ 20\ \ 10,\ 10\ \ 10\ \ 10\ ]$$

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$$F = [100, 80, 40, 20]$$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

$$[200]$$

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$$[-1\ \ -2\ \ -1,\ 0\ \ \ 0\ \ \ 0,\ \ 1\ \ 2\ \ 1\ ]$$

to:

$$[10 \;\; 10 \;\; 10, \;\; 20 \;\; 20 \;\; 20, \;\; 30 \;\; 30 \;\; 30]$$

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

$$[1 \;\; 0 \;\; 2]$$

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

$$[1 \;\; 0 \;\; 5]$$

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^{\gamma}$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, \;\; 100, \;\; 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

(a) Interpret the need for equalization.

(b) Justify use of CDF.

(c) Compute cumulative distribution.

(d) Compute new intensity levels.

(e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.
(a) Interpret noise characteristics.
(b) Justify median filter usage.
(c) Apply filter to:
$10 \quad 200 \quad 10, 200 \quad 50 \quad 200, 10 \quad 200 \quad 10$
(d) Compute center value.

(e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.
(a) Interpret smoothing objective.
(b) Justify Gaussian filter over averaging.
(c) Apply filter:
$[1 \quad 2 \quad 1, \quad 2 \quad 4 \quad 2, \quad 1 \quad 2 \quad 1]$
to matrix:
$[10 \quad 20 \quad 10, 20 \quad 40 \quad 20, 10 \quad 20 \quad 10]$
(d) Compute center pixel.

(e) Interpret smoothing effect.

16. High-Pass Filtering

An image requires sharpening.
(a) Interpret need for high-pass filtering.
(b) Justify filter selection.

(c) Apply mask:
$[-1 \quad -1 \quad -1, \ -1 \quad 8 \quad -1, \ -1 \quad -1 \quad -1]$
(d) Compute center value.

(e) Interpret edge enhancement.

17. Frequency Domain High-Pass

Given frequency components:
$F = [10, 20, 80, 100]$
(a) Interpret high-frequency importance.

(b) Justify high-pass filtering.

(c) Apply filtering (remove low frequencies).

(d) Compute output.

(e) Interpret result.

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

(a) Interpret segmentation challenge.

(b) Justify adaptive thresholding.

(c) Apply local threshold (mean = 100).

(d) Compute binary output.

(e) Interpret result.

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:
$[-1 \quad 0 \quad 1, \ -2 \quad 0 \quad 2, \ -1 \quad 0 \quad 1]$
Matrix:
$[10 \quad 20 \quad 30, \ 10 \quad 20 \quad 30, \ 10 \quad 20 \quad 30]$
(a) Interpret edge detection objective.

(b) Justify operator choice.

(c) Apply step-by-step.

(d) Compute center value.

(e) Interpret edge direction.

20. Prewitt Edge Detection

(a) Interpret Prewitt operator purpose.
(b) Justify use over Sobel.

(c) Apply kernel:
$[-1\ \ 0\ \ 1,\ \ -1\ \ \ 0\ \ \ 1,\ \ -1\ \ \ 0\ \ \ 1]$
(d) Compute output.

(e) Interpret result.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

(a) Interpret segmentation problem.

(b) Justify splitting method.

(c) Apply threshold 80.

(d) Compute regions.

(e) Interpret separation.

22. Region Merging

Regions: [45, 50], [52, 55]

(a) Interpret merging criteria.

(b) Justify merging approach.

(c) Apply threshold = 5.

(d) Compute merged region.

(e) Interpret result.

23. Edge Linking

Edges detected: [1, 0, 1, 0, 1]

(a) Interpret problem of discontinuity.

(b) Justify linking method.

(c) Apply linking step.

(d) Compute result.

(e) Interpret continuity.

24. Low-Pass Filtering (Frequency)

Given $F = [200, 150, 50, 20]$

(a) Interpret smoothing requirement.

(b) Justify LPF selection.

(c) Apply filter.

(d) Compute output.

(e) Interpret effect.

25. Laplacian Sharpening (Variant)

Mask:

$$\begin{matrix} 0 & -1 & 0, \\ -1 & 5 & -1, \\ 0 & -1 & 0 \end{matrix}$$

(a) Interpret sharpening need.

(b) Justify mask.

(c) Apply to matrix.

(d) Compute center.

(e) Interpret result.

26. Histogram Matching

(a) Interpret need for matching.

(b) Justify method.

(c) Map values [10, 50, 100].

(d) Compute new values.

(e) Interpret.

27. Noise Reduction Comparison

(a) Interpret noise type.

(b) Compare averaging vs median.

(c) Apply both.

(d) Compute outputs.

(e) Interpret differences.

28. Combined Filtering

(a) Interpret need for smoothing + sharpening.

(b) Justify sequence.

(c) Apply filters stepwise.

(d) Compute output.

(e) Interpret improvement.

29. Multi-Threshold Segmentation

Pixels: [20, 60, 120, 180]

(a) Interpret multi-level segmentation.

(b) Justify thresholds (50,100).

(c) Apply segmentation.

(d) Compute classes.

(e) Interpret.

30. Boundary Detection

(a) Interpret boundary extraction.

(b) Justify method.

(c) Apply edge + linking.

(d) Compute result.

(e) Interpret boundaries.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2_AT01

- Analytical Problem Solving

G. Madhan kumar
192472328
ITA0504
Computer Vision

1. Grey Level Transformation:

(a) Purpose : Grey level transformation improves image brightness and contrast by modifying pixel intensity values.

(b) Negative transformation formula:

$$S = L - 1 - r$$

For an 8-bit image:

$$S = 255 - r$$

Cannot compute because the pixel values are missing.

19/100

2. Log Transformation

(a) Log transformation enhances dark pixels by expanding low-intensity values and compressing high-intensity values.

(b) Formula:

$$S = c \log(1 + r)$$

3. Histogram Processing

Histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels = 16

Cumulative Distribution

Intensity	Frequency	CDF
0	2	2
1	2	4
2	4	8
3	8	16

For a 3-bit image :

$$L = 8$$

New intensity:

$$S_K = \frac{CDF}{16} \times 7$$

Intensity	CDF	New Level
0	2	1
1	4	2
2	8	4
3	16	7

4. Spatial Filtering:

(a) Used to reduce image noise and smooth intensity variations

(b) Cannot compute because the 3x3 matrix is missing

Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Sum

$10+20+30+20+30+40+30+40+50 = 270$

Average

$\frac{270}{9} = 3$

output

30

5. Spatial Filtering (sharpening)

(a) Sharpening enhances edges and fine details, making an image clear.

(b) Using the given laplacion mask:

$\boxed{\text{center output} = 40}$

6. (a) Low-pass filters

High-pass filters

(b) Given,

$F = [100, 80, 40, 20]$

Retaining first two components

$[100, 80, 0, 0]$

7. Thresholding:

(a) Global thresholding uses a single threshold value for the entire image to separate objects from the background.

(b) Given,

$T = 100, \quad r = 200$

$\because \; 200 > 100$

$\boxed{\text{output} = 1}$

7. Contrast stretching.

8. Edge Detection (Horizontal sobel)

(a) Edge Detection identifies object boundaries where pixel intensity changes rapidly.

(b) Sobel mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

image $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

Calculation

$= (-1\times10) + (-2\times10) + (-1\times10) + (0\times20) + (0\times20) + (0\times20) + (1\times30) + (2\times30) + (1\times30)$

$= -10 - 20 - 10 + 30 + 60 + 30$

$= 80$

$\boxed{\text{Gradient} = 80}$

9. Region - Based segmentation:

(a) Region growing starts from a seed pixel and adds neighboring pixels.

(b) Seed = 50, Threshold = 5

pixel in range 45 - 55 belong to the region.

10. Combined ~~Enhancement~~ & segmentation.

(a) Smoothing removes noise before segmentations, improving accuracy.

(b) Mean = 48, Threshold = 50

$48 < 50 \Rightarrow$ Output $= 0$ Background.

11. Power - Law (Gamma) Transformation.

(a) Gamma Correction adjusts image brightness

(b) for $r = 0.5$

(d) & (c) $S = \sqrt{r} \Rightarrow \sqrt{25} = 5$, $\sqrt{100} = 10$, $\sqrt{225} = 15$

[5, 10, 15]

(e) Brightness of dark region is enhanced

12. Contrast Stretching:

(a) Improves image contrast

(b) Maps intensity from 50-150 to 0-255

C,d) 50-0

100-128

150-255

e) Contrast become higher

13. Histogram Equalization:

(a) Improve image contrael

(b) CDF redistributes intensity values

(c) Cummutative frequencies : 4, 10, 20, 26

(d) Intensity levels : 2, 6, 12, 15

(e) Contrast is enhanced.

14. (a) Salt - and - pepper noise apperar as black and white dots

(b) Median filters removes this noise while preserving edges

(c) & (d) values:

10, 200, 10, 200, 50, 200, 10, 200, 10

Sorted:

10, 10, 10, 10, 50, 200, 200, 200, 200

Median = 50

(e) Noise is effively removed

15. (a) Reducers image noise

(b) Gaussion filter gives smoother result then averaging

(c) & (d) weighted sum = 620, kernel sum = 16

output = 640/16 = 40

(e) Image becomes smooth with less blur.

(16) (a) Enhances edges

(b) High-pass filter highlights fine details

c (&d) Output = 8 × 20 − 8 × 10 = 80

e) Edge become sharper.

(17.) (a) High frequency Domain High-pass

(b) High pass filtering sharpens images

c) & d) Removes low frequency

[0, 0, 80, 100]

e) Edges are enhanced

(18) (a) Handles uneven illumination

(b) Use local threshold instead of one global threshold.

c & d) Threshold = 100

60 − 0

80 − 0

~~120 − 1~~

140 − 1 [0, 0, 1, 1]

e) Better segmentation.

(19) a) Detects verticals edges

b) Sobel provides good edge detection.

c & d) Output = 80

e) Vertical edge detected

20. a) Detects image edges

b) Simpler than Sobel

c,d) Output : 60

e) Vertical edge detected

21. Region splitting

(a) Splits an image into regions based on intensity differences.

(b) Used when pixels are not simillar.

(c) Threshold = 80

(d) Regions :

- Region 1 : [40, 45]
- Region 2 : [100, 105]

22. Region Mergin

(a) Combines similar neighboring regions.

(b) Reduce over-segmentation

(c) Threshold = 5

(d) Difference = 52 - 50 = 2 < 5

Merged Region :

[45, 50, 22, 55]

23. Edge Linking

(a) Connects broken edges

(b) Produces continuous boundaries

(c) Link edge gaps

(d) output :

[1, 1, 1, 1, 1]

24) Low-pass filtering

(a) Removes high-frequency noise

(b) Smooths the image

(c) $F = [200, 150, 50, 20]$

Retain first two components

(d) Output

$[200, 150, 0, 0]$

25) Laplacian sharpening

(a) Enhances edges.

(b) Sharpnes image details.

(c) Apply mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

(d) Produces a sharper image

26) Histogram Matching

(a) ~~Matches~~ an image histogram to a reference histogram.

(b) improves image appearence. and consistency

(c) Map values: $[10, 50, 100]$

(d) Output image has a similar intensity distribution to the reference.

28) Combined Filtering

(a) Removes noise and sharpnes the image

(b) Apply smoothing first, then sharpening

(c) steps:

1. Gaussian
2. High-pass

(d) Produces a smooth and sharp image

1) Multi - Threshold Segmentation

(a) Divides an image into multiple regions

(b) Threshold : 50 and 100.

(c) Pixels : [20, 60, 120, 180]

(d) Classes :

20 → class 1   120 → class 3

60 → class 2   180 → class 3

Output : [C_1, C_2, C_3, C_3]

30) Boundary Detection

(a) Extracts object boundaries

(b) Uses edge detection and edge linking

(c) steps :

1. Detect edges
2. Link connected edges.